Md. Rawnak Ahasan

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PROFESSIONAL SUMMARY

Computer Science and Engineering student at BRAC University with hands-on experience in full-stack development, cybersecurity, machine learning, and robotics. Experienced in building security-focused web applications, ML pipelines, and robotics systems, with interests in AI, intelligent systems, and research-driven engineering.

EDUCATION

BRAC University | Merul Badda, Dhaka
B.S.C., Computer Science and Engineering | May 2023 – May 2027

TECHNICAL SKILLS

- **Languages:** Python, C, C++, JavaScript, TypeScript, PHP, SQL, HTML, CSS, Bash
- **Frontend:** React, Next.js, Vue.js, TypeScript, Tailwind CSS
- **Backend:** Node.js, Express.js, Laravel, RESTful APIs
- **Databases:** MongoDB, MySQL, Google Sheets API
- **Security:** RSA, ECC, PBKDF2, HMAC, Authentication, RBAC
- **AI / Machine Learning:** Machine Learning, Data Preprocessing, EDA, Classification, Model Evaluation
- **Tools & Infrastructure:** Git, GitHub, ROS2, Canva, TurtleSim, Docker, Linux

PROJECTS

1. CampusNest: Student Housing & Roommate Finder | Full-Stack & Security Developer

- Developed a full-stack student housing and roommate-matching platform using React, Tailwind CSS, Express.js, and MongoDB, with RESTful API architecture.
- Implemented a custom cryptographic security layer featuring RSA-512, ECC secp256k1/EC-ElGamal, PBKDF2-HMAC-SHA256, HMAC-SHA256, and CBC-MAC.
- Built 2FA, role-based access control, encrypted data storage, secure session management, key management, and cryptographic key rotation.
- Developed a multi-dimensional roommate compatibility algorithm and encrypted peer-to-peer messaging system.

2. End-to-End Machine Learning Pipeline for Subscription Attrition - Contributor

- Developed a customer churn prediction pipeline using Python, performing extensive EDA on a 1,500-record dataset to identify key behavioral drivers of customer retention.
- Optimized data quality and feature engineering by implementing mean imputation for missing values, Label/One-Hot encoding for categorical variables, and correlation-based feature selection to reduce multicollinearity.

3. Emergency BD: Real-Time Crime Reporting & Visualization Platform | Technical Lead

- Developed a web-based platform for **GPS-based crime reporting, incident visualization, and administrative moderation**.
- Implemented an administrative dashboard for **user management, report moderation, account suspension, and incident verification**.
- Designed a token-based credit mechanism to discourage spam reporting and improve platform data integrity.

4. Nova Siege - 3D Space Shooter | Contributor

- Developed a 3D space shooter using **Python and PyOpenGL**.
- Implemented custom 3D rendering, weapon mechanics, physics, and local multiplayer functionality.

5. Modern High-Performance Blog Engine(SheezBlog) - Personal Project.

- Developed a full-stack blogging platform using Next.js, React, TypeScript, and Google Sheets API as a relational database.
- Implemented RBAC, hybrid authentication, OTP verification, authorship workflows, and admin security audit trails.

6. Algae Bio-Reactor for Industrial Pollution Control -Technical Lead

- Designed a Chlorella vulgaris–based algae bioreactor to sequester CO_2 and industrial pollutants, improving air quality in Dhaka's industrial zones and BRAC University.
- Achieved 2nd position in the national round of an international science fair.

WORK EXPERIENCE

Octobrain - Robotics Trainer. *(2026 - Present)*

- Conduct hands-on robotics and STEM learning sessions for students.
- Teach basic robotics, programming, and problem-solving concepts.

Urbor AgroFintech - Head of Human Resources and Communications. *(2025 - 2026)*

- Directed HR operations and optimized employee engagement through structured KPI tracking.
- Managed investor relations, corporate communications, and compiled financial records to support strategic decisions.

Team Bionergy - Co-Lead (Technical team). *(2025 - Present)*

- Worked on development of the machine.
- Developed circuits and integration.

Team Aronnyok - Chief Technical Lead. *(2024 - Present)*

- Worked on Research and development of the product.
- Lead the team to win various competitions.

VOLUNTEERING

BRAC University Computer Club - Assistant Director, Finance Department. *(2024 - Present)*

- Participated and directed in competitive events, projects.
- Assisting in organizing activities and fostering collaboration within the club.

BRAC University History Club - Secretary, Marketing & Public Relations Department. *(2025 - Present)*

- Engaged in historical discussions, events, and research activities.
- Promoted historical awareness.

Robotics Club of BRAC University (ROBU) - Executive, Strategic Planning Department. *(2024 - Present)*

- Engaged in hands-on robotics projects and workshops.
- Contributed to the design and development of innovative robotic solutions.

BRAC University Electrical and Electronics Club - Apprentice, Research & Development Department. *(2025-Present)*

- Contributed to research initiatives.
- Managed and volunteered in development activities.

ACHIEVEMENTS & CERTIFICATIONS

- Semi-Finalist - **Undergrad Startup Challenge 2025, BRAC University Entrepreneurship Development Forum (BUEDF).**
- Silver Medalist **- University & Life Science Category, ISIF Bangladesh Round, 2025.**
- Bronze Medalist - **Innovation World Cup (IWC) Bangladesh Round, Malaysia Innovation, Invention & Creativity Association. 2026.**
- 2nd Runner-Up **- IPE Carnivesta, AUST IPE Society, 2025.**
- Certificate of Completion **- Introduction to Cyber Security, Cisco Networking Academy, 2026.**
- Certificate of Completion **- Introduction to Data Analytics, Bohubrihi, 2024.**
- Certificate of Completion **- Introduction to Python Programming, Bohubrihi, 2024.**
- Certificate of Completion **- Pathway to Software Engineering, Bohubrihi, 2024.**
- Certificate of Participation - **Battle of insights, Data Solution-360, 2024.**
- Certificate of Appreciation **- Administration Department, Outlive, 2021.**

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